



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE – Raigad - 402103

End semester Examination (Summer) May 2018-19

SEM: IV

Branch: B. Pharmacy

Subject with Subject Code: Pharmacology I (BP404T)

Marks: 75

Date: 03/06/2019

Time: 10.30am to 1.30 pm (3 Hrs.)

Instructions: - 1) All questions are compulsory.

2) Figures to the right indicate full marks

3) Draw figures and charts wherever necessary

Q.1. Attempt the following multiple choice question (1×20)

(20 Marks)

A) What does “pharmacokinetics” include?

- | | |
|--------------------------------|------------------------------|
| a) Localization of drug action | b) Mechanisms of drug action |
| c) Excretion of substances | d) Interaction of substances |

B) What does the term “bioavailability” mean?

- Plasma protein binding degree of substance
- Permeability through the brain-blood barrier
- Fraction of an uncharged drug reaching the systemic circulation following any route administration
- Amount of a substance in urine relative to the initial dose

C) Tolerance develops because of:

- | | |
|--------------------------|------------------------------|
| a) Diminished absorption | b) Rapid excretion of a drug |
| c) Both of the above | d) None of the above |

D) Pharmacodynamics involves the study of following EXCEPT:

- | | |
|--|---|
| a) Biological and therapeutic effects of drugs | b) Absorption and distribution of drugs |
| c) Mechanisms of drug action | d) Drug interactions |

E) Indicate the local anesthetic agent, which has a shorter duration of action:

- | | | | |
|--------------|-------------|----------------|----------------|
| a) Lidocaine | b) Procaine | c) Bupivacaine | d) Ropivacaine |
|--------------|-------------|----------------|----------------|

F) Indicate the location of M2 cholinoreceptor type:

- | | | | |
|----------|-----------|------------------|----------------|
| a) Heart | b) Glands | c) Smooth muscle | d) Endothelium |
|----------|-----------|------------------|----------------|

G) Catecholamine includes following EXCEPT:

- | | | | |
|--------------|----------------|-----------------|-------------------|
| a) Ephedrine | b) Epinephrine | c) Isoprenaline | d) Norepinephrine |
|--------------|----------------|-----------------|-------------------|

H) A bronchial smooth muscle contains:

- | | | | |
|-------------------|-------------------|--------------------|--------------------|
| a) Alfa1 receptor | b) Alfa2 receptor | c) Beta 1 receptor | d) Beta 2 receptor |
|-------------------|-------------------|--------------------|--------------------|

I) Propranolol has all of the following cardiovascular effects EXCEPT:

- a) It decreases cardiac work and oxygen demand
- b) It reduces blood flow to the brain
- c) It inhibits the renin secretion
- d) It increases the 'atrioventricular nodal refractory period

J) Hypnotic drugs are used to treat:

- a) Psychosis b) Sleep disorders c) Narcolepsy d) Parkinsonian disorders

K) Which of the following drugs is used in the treatment of Parkinsonian disorders?

- a) Phenytoin b) Selegiline c) Haloperidol d) Fluoxetine

L) Alcohol may cause:

- a) CNS depression b) Vasodilatation c) Hypoglycemia d) All of the above

M) Which of the following analgesics is a phenanthrene derivative?

- a) Fentanyl b) Morphine c) Methadone d) Pentazocine

N) Neuroleptics are used to treat:

- a) Neurosis b) Narcolepsy c) Psychosis d) Parkinsonian disorders

O) Monoamine Oxidase A:

- a) Is responsible for norepinephrine, serotonin, and tyramine metabolism
- b) Is more selective for dopamine
- c) Metabolizes norepinephrine and dopamine
- d) Deaminates dopamine and serotonin

P) Which of the following benzodiazepines has the shortest duration of action?

- a) Triazolam b) Clorazepate c) Prazepam d) Clordiazepoxide

Q) Indicate the nootropic agent:

- a) Sydnocarb b) Eleuterococci extract c) Fluoxetine d) Piracetam

R) Tolerance is associated with:

- a) An ability to compensate for the drug effect
- b) Increased disposition of the drug after chronic use
- c) Compensatory changes in receptors, effector enzymes
- d) All of the above

S) Which route of drug administration is most likely to lead to the first-pass effect?

- a) Sublingual b) Oral c) Intravenous d) Intramuscular

T) Half-life ($t_{1/2}$) doesn't depend on:

- a) Biotransformation b) Time of drug absorption
c) Concentration of a drug in plasma d) Rate of drug elimination

Q.2. Solve any Two of the following (2×10)

(20 Marks)

- A) Enlist & explain in detail factors modifying drug action.
B) Define Local Anesthetics & give its classification, Pharmacological action, uses.
C) Explain in detail synthesis, storage, release & termination of following-
1) Dopamine
2) Serotonin

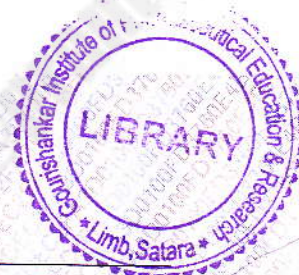
Q.3. Solve any Seven of the following (7×05)

(35 Marks)

- A) Define Clinical trials & explain phases of Clinical trials.
B) Define-
1) Pharmacovigilance 2) Pharmacokinetics 3) Agonist
4) Toxicology 5) Parasympathomimetics
C) Explain in detail synthesis, storage, release & binding of Acetylcholine.
D) Explain in detail synthesis, storage, release of GABA & enlist function of GABA.
E) Write the classification of Narcotic drugs & explain MOA, Pharmacokinetics, Side effect of same.
F) Write the Mechanism of Action of following-
1) Atenolol 2) Nitrous oxide
G) Enlist various routes of Administration & give advantages disadvantages of any two routes.
H) Write note on Receptor.
I) Explain mechanism of action & Pharmacological action of Adrenaline.

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE - RAIGAD - 402 103

Winter Semester Examination -Dec - 2019



Course: B. Pharm.

Subject: Pharmacology I (BP404T)

Date: 11/12/2019

Semester: IV

Marks: 75

Duration: 3hrs

Instructions: i) All questions are compulsory

ii) Figures to the right indicate full marks

iii) Draw the diagrams or flow charts wherever necessary.

Q. No.1 Attempt the following questions (All Questions Compulsory) (20 Marks)

1. _____ disease may be induced by drug
 - A. Peptic Ulcer
 - B. Hepatitis
 - C. Parkinsonism
 - D. All
2. How many helix are present in G-protein receptor?
 - A. six helix
 - B. seven helix
 - C. three helix
 - D. four helix
3. Which of the following agent is precursor of dopamine
 - A. Bromocriptine
 - B. Levodopa
 - C. Selegiline
 - D. None of these
4. Alcohol shows synergistic effect with _____
 - A. Imipramine
 - B. Diazepam
 - C. Barbiturates
 - D. All of these
5. Which one of the following categories of adrenergic drug used in bronchial asthma?
 - A. Selective β_1 agonist
 - B. Selective β_2 agonist
 - C. Selective β_1 antagonist
 - D. Selective β_2 antagonist
6. First pass metabolism occurs in _____
 - A. Stomach
 - B. Lung
 - C. Pancreas
 - D. Liver
7. Indicate competitive antagonist of BZ receptors
 - A. Flumazenil
 - B. Picrotoxin
 - C. Zolpidem
 - D. Temazepam
8. No surgical operation should be performed during _____ stage of anesthesia.
 - A. I

- B. II
C. III
D. IV
9. Pilocarpine is commonly used in treatment _____
A. Myasthenia gravis
B. Mushroom poisoning
C. Anticholinesterase poisoning
D. Glaucoma
10. There are _____ types of muscarinic receptors
A. 2
B. 3
C. 4
D. 5
11. Phenytoin acts on _____
A. Na⁺ channel
B. Ca⁺⁺ channel
C. Both a and b
D. None of these
12. Metabolic transformation (phase I) is
A. Acetylation & methylation of substances
B. Transformation of substance due to oxidation, reduction or hydrolysis
C. Glucuronide conjugation
D. Binding to plasma protein
13. Anticholinergic effect on human eye is _____
A. Cycloplegia
B. Mydriasis
C. ↑ IOP
D. All
14. _____ Anticholinesterase useful in Alzheimer's disease
A. Arecolin
B. Donepezil
C. Isoproterenol
D. Clioquinol
15. What drug does to the body is _____
A. Pharmacokinetics
B. Pharmacodynamics
C. Pharmacotherapeutics
D. Pharmacogenomics
16. What is the reason of complicated penetration of some drugs through brain- blood barrier?
A. High lipid solubility of drug
B. Meningitis
C. Absence of pores in the brain capillary endothelium
D. High endocytosis degree in brain capillary
17. Phenytoin is used in treatment of:
A. Petit mal epilepsy
B. Grand mal epilepsy
C. Myoclonic Seizure
D. All of the above
18. Atropine causes

- A. Bradycardia & Bronchoconstriction
 - B. Tachycardia & Bronchodilation
 - C. Hypotension & Bronchoconstriction
 - D. Hypotension & Bronchodilation
19. _____ is not produced by predictable adverse drug reaction.

- A. Intolerance
 - B. Rebound Response
 - C. Secondary pharmacological response
 - D. None
20. Which one of the following example of ion channel receptor?
- A. Serotonin
 - B. Nicotinic
 - C. Dopaminergic
 - D. Muscarinic

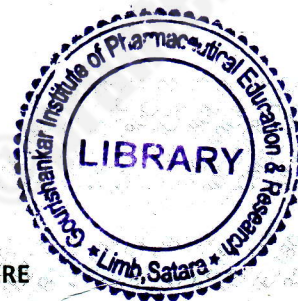
Q. No.2 Attempt any TWO questions of the following: (20 Marks)

- A] Explain in detail synthesis, storage, release, uptake & metabolism of catecholamines & explain pharmacological action of adrenaline.
- B] Discuss various mechanisms of drug action and explain various receptors.
- C] What are general anesthetics? Explain phases of anesthesia, Add note on inhalation anesthetics.

Q. No.3 Attempt any SEVEN questions of the following: (35 Marks)

- A] Give advantages and disadvantages of parenteral route.
- B] Define sedative hypnotic and describe pharmacology of Benzodiazepines.
- C] Define and classify antidepressant drugs.
- D] Discuss five therapeutic uses of anticholinergic drugs.
- E] What is Parkinsons Disease? Classify drugs used in Parkinsons Disease. Why combination of Levodopa and Carbidopa is prescribed in Parkinsons Disease?
- F] Enlist factors affecting drug absorption and explain any two in detail.
- G] Define the terms
 - i) Pharmacotherapeutics
 - ii) Teratogenicity
 - iii) Bioavailability
 - iv) Synergism
 - v) Toxicology
- H] Explain basic steps involved in Neurohumoral transmission.
- I] Define adverse drug reaction & explain types of adverse drug reaction.

----- PAPER END -----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Summer 2022

Course: B. Pharmacy

Sem: IV

Subject Name: Pharmacology - I

Subject Code: BP404T

Max Marks: 75

Date: 06/09/2022

Duration: 3.45 Hr.

Instructions –

1. All questions are compulsory
2. Answers to MCQs should be written in full sentences
3. Draw diagrams / figures wherever necessary
4. Figures to right indicate full marks

Q. 1. Multiple Choice Questions (MCQs) = 20 x 1 = 20 (All the questions are compulsory)

- i) Pharmacokinetics involves study of the _____ of a drug.
- | | |
|-----------------|------------------------|
| a) Side effects | b) Mechanism of action |
| c) Distribution | d) Therapeutic effects |
- ii) By which route of administration, the extensive first-pass metabolism of the drugs occurs?
- | | |
|------------------|----------------|
| a) Oral | b) Intravenous |
| c) Intramuscular | d) Inhalation |
- iii) Which is the fastest route of administration in terms of the effect of a drug?
- | | |
|------------------|-----------------|
| a) Intramuscular | b) Subcutaneous |
| c) Intravenous | d) Intradermal |
- iv) Which organ is involved in the excretion of the drugs?
- | | |
|----------|-----------|
| a) Heart | b) Spleen |
| c) Brain | d) Kidney |
- v) Which statement is CORRECT for the agonist?
- | |
|---|
| a) It binds to a receptor and produces the biological effect |
| b) It binds to a receptor and does not produce the biological effect |
| c) It binds to a receptor and produces the biological effect opposite to that of the natural ligand |
| d) None of the above |
- vi) On repeated use if the effect of a drug decreases progressively, the mechanism is called _____
- | | |
|----------------|--------------|
| a) Antagonism | b) Tolerance |
| c) Anaphylaxis | d) Synergism |

vii) The ability of a drug to cause abnormalities in the developing fetus when administered to a pregnant woman is called _____.

- a) Teratogenicity
- b) Tachyphylaxis
- c) Superinfection
- d) Anaphylaxis

viii) Clinical trials are carried out in _____.

- a) Rats
- b) Mice
- c) Rabbits
- d) Humans

ix) Which is an adrenergic receptor?

- a) Nicotinic receptor
- b) Mu receptor
- c) Beta receptor
- d) GABA-A receptor

x) Which drug is used for the treatment of myasthenia gravis?

- a) Neostigmine
- b) Atropine
- c) Donepezil
- d) Rivastigmine

xi) Which of the following is NOT the effect of acetylcholine?

- a) Miosis
- b) Increase in rate and force of contraction of the heart
- c) Bronchoconstriction
- d) Gastric acid secretion

xii) By inhibiting which ion channels local anesthetics block nerve conduction?

- a) Na^+
- b) Mg^{2+}
- c) Ca^{2+}
- d) Cl^-

xiii) Which adverse effect is associated with chronic alcoholism?

- a) Fatty liver
- b) Gastritis
- c) Impotency
- d) All of the above

xiv) Disulfiram is used for the treatment of _____ dependence.

- a) Opioid
- b) Nicotine
- c) Alcohol
- d) Benzodiazepine

xv) Sympathetic hyperactivity associated with anxiety disorders can be treated by _____.

- a) Propranolol
- b) Oxazepam
- c) Alprazolam
- d) Chlordiazepoxide

xvi) Which of the following drug is an opioid antagonist?

- a) Methadone
- b) Naloxone
- c) Ephedrine
- d) Fentanyl

xvii) Which drug is used as a first-line drug in the absence seizures?

- a) Phenytoin
- b) Carbamazepine
- c) Ethosuximide
- d) All of the above

xviii) Which is the common adverse effect of selective serotonin reuptake inhibitors (SSRIs)?

- a) Hypotension
- b) Sexual dysfunction
- c) Seizures
- d) Arrhythmia

xix) Which antiviral drug is used in the treatment of Parkinson's disease?

- a) Entacapone
- b) Amantadine
- c) Bromocriptine
- d) Selegiline

xx) Which centrally acting anticholinesterase is not approved for Alzheimer's disease?

- a) Rivastigmine
- b) Donepezil
- c) Galantamine
- d) Physostigmine

Q. 2. Long Answers) = 2 x 10 = 20 (Answer 2 out of 3)

- i) Define drug absorption. Explain the factors affecting drug absorption.
- ii) Define neurohumoral transmission. Explain different steps involved in the neurohumoral transmission of acetylcholine.
- iii) Classify antipsychotic drugs. Explain the mechanism of action, uses, and side effects of chlorpromazine.

Q. 3. Short Answers = 7 x 5 = 35 (Answer 7 out of 9)

- i) Define drug. Write different sources of drugs with suitable examples.
- ii) Write the advantages and disadvantages of the oral route of drug administration.
- iii) Classify receptors with suitable examples. Draw a schematic diagram showing G protein-coupled receptors signaling pathway.
- iv) Explain pharmacotherapy of glaucoma.
- v) Classify cholinergic receptors, and give their distribution.
- vi) Write the mechanism of action of benzodiazepines. Why benzodiazepines are preferred over barbiturates?
- vii) Describe different stages of inhalational general anesthesia.
- viii) Classify drugs used for the treatment of Alzheimer's disease. Explain the mechanism of action of memantine.
- ix) Classify antiparkinsonian drugs. Why levodopa is given in combination with carbidopa?

-----END OF THE PAPER-----



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
End Semester Examination – Winter-2022

Course: B. Pharmacy

Sem: IV

Subject Name: Pharmacology-I

Subject Code: BP404T

Max Marks: 75

Day & Date: Tuesday, 14/02/2023

Time: 02:00 pm to 05:00 pm

Instructions –

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions

10 x 2 (Answer all the questions)

1. Explain Pharmacokinetic and Pharmacodynamics.
2. What is enzyme induction phenomenon?
3. Write in brief, role and distribution of 5HT.
4. Enlist the neurotransmitter in CNS.
5. What do you mean by drug receptor interaction?
6. Write merits and demerits of rectal route of drug administration.
7. Define drug addiction and drug dependence.
8. Write in brief the drug treatment in the Alzheimer's disease.
9. What is drug synergism? Give examples.
10. What does the term "bioavailability" mean?

Q. 2. Long Answers

2 x 10 = 20 (Answer 2 out of 3)

1. Describe the mechanisms involved in drug excretion and the factors that influence it.
2. What are Parasympatholytics? Classify Parasympatholytic drugs with suitable examples and explain pharmacology of atropine.
3. Classify antipsychotic drug and explain pharmacology of chlorpromazine.

Q. 3. Short Answers

7 x 5 = 35 (Answer 7 out of 9)

1. Take brief account of adverse drug reaction.
2. Classify Anti-Parkinsonian drugs with examples. Explain the pharmacology of Levodopa.
3. Explain the JAK-STAT binding receptor transduction mechanism.
4. Write the mechanism of action and pharmacology of sodium valproate.

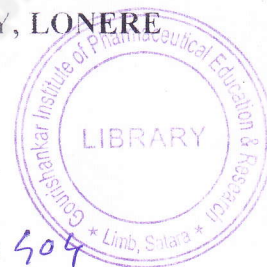
5. Classify general anesthetic with example and stages of anesthesia.
6. Discuss pharmacotherapy of Alcoholism.
7. Classify Opioid analgesics and Explain pharmacology of Morphine.
8. Explain pharmacology of MAO inhibitors.
9. Note on Clinical trials.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination –Summer 2023

Date: 17/07/2023



Course : B. Pharmacy
Subject Name : Pharmacology I

Sem: IV
Subject Code : 404
BP404T
Duration : 3 Hr.

Max Marks : 75

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2 = 20)

- i) Explain the term Therapeutic Index .
- ii) What is meant by :
 - a) Pharmacokinetics
 - b) Pharmacodynamics
- iii) Write different cholinergic receptors with their location and function.
- iv) Mention any two examples of drugs used in
 - a) Parkinsonism
 - b) Alzheimer
- v) Write uses of Atropine sulphate.
- vi) What are preanaesthetics? Write examples of drugs used as preanaesthetics.
- vii) Define with examples :
 - a) Antipsychotics
 - b) Antianxiety agents
- viii) Write any two examples of drugs used as :
 - a) Skeletal muscle relaxants
 - b) Nootropics
- ix) What is Pharmacovigilance and Enzyme Induction?
- x) Write different sources of drugs with examples.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10 = 20)

- i) State various routes of drug administration with examples.
Write down advantages and disadvantages of
 - a) Sublingual route
 - b) Oral route
- ii) Define and classify sedative hypnotics with examples.
Explain mechanism of action of Benzodiazepines.
- iii) Explain in detail factors affecting action of drug.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Define Antidepressants. Write short note on SSRIs.
- ii) Write drug therapy for Glaucoma.
- iii) Define local anaesthetics. Explain their mechanism of action.
- iv) Write short note on clinical trial.
- v) What are different types of receptors. Explain any one.
- vi) Explain synthesis, storage, metabolism and release of Acetylcholine.
- vii) Write uses of Morphine.
- viii) Write short note on Antagonism.
- ix) Write uses of α blockers.

-----END OF THE PAPER-----

Course: B. Pharmacy
Subject Name: Pharmacology-I
Max Marks: 75

Date: 09/01/2024

Sem:

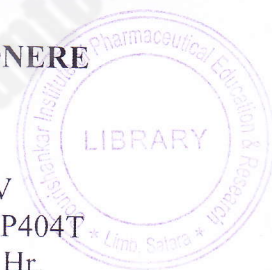
Subject Code:

Duration:

IV

BP404T

3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. The figures to the right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- i) Define prodrug. Give examples of prodrugs.
- ii) Enlist the drugs used in glaucoma.
- iii) What is the dose-response relationship? What are its advantages?
- iv) What is Drug addiction? Give examples.
- v) Name and role of excitatory neurotransmitters present in CNS.
- vi) Define Synergism Classify it with examples.
- vii) Differentiate between Enzyme Induction and Enzyme Inhibition.
- viii) Define –i) Agonist ii) Antagonist
- ix) Define –i) Sedative ii) Hypnotics
- x) Define bioavailability and clearance.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- i) Classify parasympathomimetics with examples. Write the pharmacology of Acetylcholine.
- ii) Define Receptor. Classify Receptors and explain G-Protein coupled receptors with signaling transduction mechanisms.
- iii) Define Parkinsonism. Classify antiparkinson drugs with examples. Write the mechanism of action and therapeutic uses of MAO inhibitors.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- i) Compare the Merits and Demerits of oral and Parenteral routes of administration.
- ii) Classify Sedative-Hypnotics with examples. Explain the mechanism of action, adverse effects, and uses of diazepam.
- iii) Write About the Mechanism and stages of general anesthesia.
- iv) Define Myasthenia gravis. Enlist the drugs Used in its treatment.
- v) Write about various phases of clinical Trials.
- vi) Write a note on Phase –I Biotransformation Reactions with Examples.
- vii) Discuss Pharmacokinetics drug interactions with suitable examples.
- viii) Mention Mechanism of action and uses of Local anesthetic agents.
- ix) Discuss in detail about pharmacological actions of Morphine.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Regular/Supplementary Summer Examination – 2024

Course: B. Pharmacy

Subject Name: Pharmacology I

Max Marks: 75

Date: 21-06-2024

Semester: IV

Subject Code: BP404T

Duration: 3 Hr.



Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10 X 2) = 20

- i) What are different sources of drugs? Give examples?
- ii) Define agonists and antagonists.
- iii) Define the concept of addiction, dependence, drug abuse and tolerance in pharmacology
- iv) What are the routes of drug administration?
- v) Enlist opioid analgesic agonist and antagonist.
- vi) What is idiosyncrasy and allergy?
- vii) What are the mechanisms of drug action in pharmacodynamics?
- viii) Enlist the classification of receptors based on receptor theories.
- ix) Explain the concept of therapeutic index.
- x) Briefly describe the phases of clinical trials in drug discovery.

Q.2. Long Answers (Answer 2 out of 3)

(10 X 2) = 20

- i) Explain the pharmacology of sympathomimetics and Sympatholytics.
- ii) Describe the Steps of neurohumoral transmission in the central nervous system.
- iii) Write the pharmacology of Diazepam (Include definition of sedative, hypnotic, classification, Mechanism of action, pharmacokinetics, adverse effect, BZD antagonists and ,uses)

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Discuss the drugs used in Parkinson's disease
- ii) Explain in brief drug used as 1. Hallucinogen 2. Mood Stabilizer
- iii) Write a note on G- Protein coupled receptors
- iv) Describe the mechanisms of drug interactions.
- v) Explain the pharmacology of local anesthetic agents.
- vi) What are the drugs used in the treatment of glaucoma?
- vii) Discuss the pharmacology of peripherally acting muscle relaxants.
- viii) Write a note on Nootropics
- ix) Write a note on Alcohol and disulfiram.

*** END OF THE PAPER ***